

■ PostgreSQL Database – Complete Explanation (Odoo Day 4)

Before learning Odoo development, you must understand PostgreSQL, because Odoo stores all its data in PostgreSQL.

What is PostgreSQL?

PostgreSQL is a Database Management System (DBMS).

It is used to:

- Store data
- Retrieve data
- Update data
- Delete data
- Manage millions of records safely

Think of PostgreSQL as Odoo's digital storage room.

Simple Definition

PostgreSQL is an open-source Relational Database Management System (RDBMS) that stores and manages all the data used by Odoo.

Real-Life Example

Imagine a school.

The school has:

- Students
- Teachers
- Classes
- Fees
- Attendance
- Exams

Where is all this information stored?

In registers or computers.

In Odoo, all this information is stored inside PostgreSQL.

School

Students

Teachers

Fees

Attendance

↓

PostgreSQL Database

What is a Database?

A database is an organized collection of information.

Example:

ID	Name	Age	City
1	Sugumar	48	Chennai
2	Raj	30	Madurai
3	Priya	28	Coimbatore

This table is stored inside PostgreSQL.

What is RDBMS?

RDBMS means:

Relational Database Management System

"Relational" means tables can be connected.

Example:

Student Table

Student ID	Name
------------	------

1	Sugumar
---	---------

2	Raj
---	-----

Marks Table

Student ID	Marks
------------	-------

1	95
---	----

2	88
---	----

The Student ID links the two tables.

Odoo Architecture
Browser



Odoo Server (Python)



ORM



PostgreSQL



Stores Data

Every time you click Save, data goes into PostgreSQL.

Example

Suppose you create a customer.

Name : John

Email : john@gmail.com

Click Save



Odoo ORM



PostgreSQL stores:

ID	Name	Email
1	John	john@gmail.com

What Does PostgreSQL Store?

Everything.

Example:

Customers

Products

Employees

Attendance

Invoices

Sales Orders

Purchase Orders

Suppliers

Stock

Payments

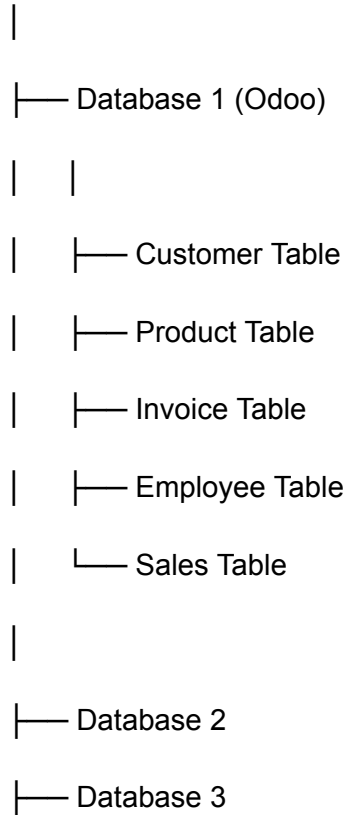
Users

Roles

Permissions

Database Hierarchy

PostgreSQL Server



One PostgreSQL server can manage many databases.

Table Example

Suppose we have Customers.

Customer Table

ID	Name	City	Phone
1	Sugumar	Chennai	9876543210
2	Raj	Madurai	9123456789

Every row is called a Record.

Every column is called a Field.

PostgreSQL Terminology

Term	Meaning
------	---------

Database Collection of tables

Table Stores records

Row One record

Column One field

Primary Key Unique ID

Foreign Key Connects two tables

Example

Student Table

Student_ID	Name
------------	------

1	Sugumar
---	---------

Attendance Table

Attendance_ID	Student_ID	Date
---------------	------------	------

100	1	01-07-2026
-----	---	------------

The Student_ID connects the two tables.

Why Odoo Uses PostgreSQL

1. Open Source

Free to use.

2. Fast

Handles millions of records efficiently.

3. Secure

Provides authentication, permissions, backups, and transaction support.

4. Reliable

Trusted by companies around the world.

5. Scalable

Works for:

Small businesses

Medium businesses

Large enterprises

How Odoo Saves Data

Suppose you create:

Product

Laptop

₹50,000

Flow:

Browser



Odoo



ORM



PostgreSQL



Product Table

The product is stored permanently.

CRUD Operations

Create

Add new data.

Add Customer



New record stored.

Read

View data.

Show all customers

↓

Records are displayed.

Update

Modify data.

Age

48

↓

49

↓

Database updates the record.

Delete

Remove data.

Delete Customer

↓

Record removed from the database.

Relationship with ORM

Without ORM:

Python

↓

SQL

↓

PostgreSQL

With Odoo ORM:

Python

↓

ORM

↓

PostgreSQL

ORM automatically generates SQL.

Real Odoo Example

When you install the Sales app:

PostgreSQL creates tables like:

sale_order

sale_order_line

product_template

res_partner

res_users

These tables store all the data used by the Sales module.

PostgreSQL in Your Chromebook

When you installed Odoo on Day 3, you also installed PostgreSQL.

You can check if it's running with:

```
sudo systemctl status postgresql
```

Or connect to it:

```
sudo -u postgres psql
```

Inside PostgreSQL, list databases:

```
\l
```

You might see:

```
postgres
```

```
template0
```

```
template1
```

```
odoo18
```

Connect to your Odoo database:

```
\c odoo18
```

List all tables:

```
\dt
```

You'll see tables such as:

```
res_partner
```

```
res_users
```

```
product_template
```

```
sale_order
```

```
account_move
```

Exit PostgreSQL:

```
\q
```

PostgreSQL vs MySQL

Feature	PostgreSQL	MySQL
---------	------------	-------

Open Source ☒ ☒
 Used by Odoo ☒ Yes ☒ No
 Advanced SQL ☒ Good
 Complex Queries Excellent Good
 Reliability Excellent Very Good
 Enterprise Features Excellent Good
 Interview Questions
 Q1. What is PostgreSQL?

Answer: PostgreSQL is an open-source Relational Database Management System (RDBMS) used by Odoo to store and manage all application data.

Q2. Why does Odoo use PostgreSQL?

Answer: Because it is reliable, secure, scalable, and supports advanced relational database features.

Q3. What does PostgreSQL store in Odoo?

Answer: It stores all business data, including customers, products, employees, invoices, sales orders, inventory, users, and application settings.

Q4. What is the difference between PostgreSQL and ORM?

PostgreSQL	ORM
Database where data is stored	Layer that communicates with the database
Stores tables and records	Converts Python code into SQL queries
Manages data permanently	Makes database operations easier for developers

Easy Way to Remember

Think of Odoo like a company office:

-  User = Employee using Odoo
-  Browser = Reception desk
-  Odoo Server (Python) = Office staff processing requests
-  ORM = Translator between Python and the database
-  PostgreSQL = The secure records room where all company information is permanently stored

When you click Save, the request travels through the server and ORM, and the data is finally stored in PostgreSQL.

When you open the record again, PostgreSQL sends the stored data back through ORM and Odoo to your browser.

